

Baddies and calls the `collide()` method of each. If there was an intersection (overlap) with any Baddie, then we report that back to the main code, which ends the game.

File: Dodger/Goodies.py

The `GoodieMgr` and `Goodie` classes are very similar to the `BaddieMgr` and `Baddie` classes. The `Goodie` manager is an object manager object that maintains a list of Goodies. The difference from the Baddie manager is that it will randomly place a Goodie at either the left edge of the window (in which case it moves right) or the right edge (so it moves left). It also creates new Goodies after a randomized number of frames. When the Player intersects with a Goodie, the user is rewarded with 25 points. The `update()` method of the `Goodie` manager uses the technique described in the previous section: it makes a copy of the Goodies list and iterates through the copy.

File: Dodger/Player.py

The `Player` class, shown in [Listing 16-10](#), manages the Player icon and keeps track of where it should appear in the game window.

```
# Player class
--- snip imports ---

class Player():
    ❶ def __init__(self, window):
        self.window = window
        self.image = pygwidgets.Image(window,
                                         (-100, -100), 'images/player.png')
        playerRect = self.image.getRect()
        self.maxX = WINDOW_WIDTH - playerRect.width
        self.maxY = GAME_HEIGHT - playerRect.height

        # Every frame, move the Player icon to the mouse position
        # Limits the x- and y-coordinates to the game area of the window
    ❷ def update(self, x, y):
        if x < 0:
            x = 0
        elif x > self.maxX:
            x = self.maxX
        if y < 0:
```